Assignment 3 杜乘旋 12209095

1. (i) $A=\begin{bmatrix}1&0&-2\\-3&1&4\\2&-3&4\end{bmatrix}$ $\left[\begin{array}{ccc|ccc}1&0&-2&1&0&0\\-3&1&4&0&1&0\\2&-3&4&0&0&1\end{array}\right]\to\left[\begin{array}{ccc|ccc}1&0&-2&1&0&0\\0&1&-2&3&1&0\\0&-3&8&-2&0&1\end{array}\right]\to\left[\begin{array}{ccc|ccc}1&0&-2&1&0&0\\0&1&-2&3&1&0\\0&0&2&7&3&1\end{array}\right]\to\left[\begin{array}{ccc|ccc}1&0&0&8&3&1\\0&1&0&10&4&1\\0&0&1&\frac{7}{2}&\frac{3}{2}&\frac{1}{2}\end{array}\right]$

$\therefore A^{-1}=\begin{bmatrix}8&3&1\\10&4&1\\\frac{7}{2}&\frac{3}{2}&\frac{1}{2}\end{bmatrix}$

(ii) $A=\begin{bmatrix}1&-2&1\\4&-7&3\\-2&6&-4\end{bmatrix}$ $|A|=28+12+24-14-32-18=0$ $\therefore$ A is singular

2. (i) $A=\begin{bmatrix}1&3\\2&1\end{bmatrix}$ $B=\begin{bmatrix}-1&-3\\2&1\end{bmatrix}$

$A^{-1}=\frac{1}{5}\begin{bmatrix}1&-3\\-2&1\end{bmatrix}$ $B^{-1}=\frac{1}{5}\begin{bmatrix}1&3\\-2&-1\end{bmatrix}$ $\therefore$ A,B is invertible

$A+B=\begin{bmatrix}0&0\\4&2\end{bmatrix}$ $|A+B|=0$ is not invertible

(ii) $A=\begin{bmatrix}0&0\\2&1\end{bmatrix}$ $B=\begin{bmatrix}1&3\\0&0\end{bmatrix}$

$|A|=0$, $|B|=0$ $\therefore$ A, B is singular matrices

$A+B=\begin{bmatrix}1&3\\2&1\end{bmatrix}$ $(A+B)^{-1}=-\frac{1}{5}\begin{bmatrix}1&-3\\-2&1\end{bmatrix}$ $\therefore$ A+B is invertible

3. $\begin{bmatrix}4&-1&-1&-1\\-1&4&-1&-1\\-1&-1&4&-1\\-1&-1&-1&4\end{bmatrix}\begin{bmatrix}a&b&b&b\\b&a&b&b\\b&b&a&b\\b&b&b&a\end{bmatrix}=\begin{bmatrix}c&d&d&d\\d&c&d&d\\d&d&c&d\\d&d&d&c\end{bmatrix}=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}$ $\therefore c=1$ $d=0$

$c=4a-3b$ $d=2b-a$

$\therefore\begin{cases}4a-3b=1\\2b-a=0\end{cases}$ $\therefore\begin{cases}a=\frac{2}{5}\\b=\frac{1}{5}\end{cases}$

4. $A=\begin{bmatrix}a&b&b\\a&a&b\\a&a&a\end{bmatrix}$ $|A|=a^3+a^2b+b^2a-3a^2b=a^3+b^2a-2a^2b$ $\because a\neq b$, $a\neq 0$ $\therefore |A|\neq 0$

$A^{-1}=\frac{1}{a^3+b^2a-2a^2b}\begin{bmatrix}a^2-ab&0&b-ab\\ab-a^2&a^2-ab&0\\0&ab-a^2&a^2-ab\end{bmatrix}$

$C=\begin{bmatrix}2&c&c\\c&c&c\\8&7&c\end{bmatrix}$ $|C|=2c^2+8c^2+7c^2-8c^2-c^3-14c=9c^2-c^3-14c$

$\because C$ is not invertible $\therefore |C|=0$ $\therefore c(9c-c^2-14)=c(-c+2)(c-7)=0$

$\therefore c=0$ or 2 or 7

5. prove: $(I-A)(I+A+A^2+\cdots+A^{N-1})=I+A+A^2+\cdots+A^{N-1}-A-A^2-A^3-\cdots-A^N=I-A^N=I$

$(I+A+A^2+\cdots+A^{N-1})(I-A)=I+A+A^2+\cdots+A^{N-1}-A-A^2-\cdots-A^{N-1}-A^N=I-A^N=I$

$\therefore (I-A)^{-1}=I+A+A^2+\cdots+A^{N-1}$

6. (i) $A=LDU \quad A=L_1D_1U_1$

$\therefore LDU=L_1D_1U_1 \quad \therefore L_1^{-1}LD=D_1U_1U^{-1}$

(Lower triangular) $D=D_1$ (Upper triangular)

$\because$ LHS=RHS $\therefore$ both sides are diagonal

$L_1^{-1}L=I=U_1U^{-1}$

$\therefore D=D_1, \quad L=L_1, U=U_1$

(ii) $LU=L_1U_1 \quad \therefore L_1^{-1}LUU^{-1}=L_1^{-1}L_1U_1U^{-1}$

$\therefore L_1^{-1}L=U_1U^{-1}$ Lower triangular = Upper triangular

$\therefore L_1^{-1}L=I=U_1U^{-1} \quad \therefore L_1=L, U_1=U$

(iii) PA and P'A both represents the factorization of the same matrix A.

It follows $LDU=L_1D_1U_1$

$\therefore P=P_1 \quad LDU=L_1D_1U_1$

$\therefore L_1^{-1}LD=D_1U_1U^{-1}$

(Lower triangular) $D=D_1$ (upper triangular)

since both sides are equal

it follows that both sides are diagonal

$\therefore D=D_1, \quad L_1^{-1}L=U_1U^{-1}=I$

$\therefore D=D_1, L=L_1, U=U_1$

7. (i) $A=\begin{bmatrix}1&1&0\\1&2&1\\0&1&2\end{bmatrix} \xrightarrow{R_2-R_1} \begin{bmatrix}1&1&0\\0&1&1\\0&1&2\end{bmatrix} \xrightarrow{R_3-R_2} \begin{bmatrix}1&1&0\\0&1&1\\0&0&1\end{bmatrix} \quad \therefore U=\begin{bmatrix}1&1&0\\0&1&1\\0&0&1\end{bmatrix}$

$\therefore L=\begin{bmatrix}1&0&0\\1&1&0\\0&1&1\end{bmatrix} \quad D=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

$A=\begin{bmatrix}1&0&0\\1&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}1&1&0\\0&1&1\\0&0&1\end{bmatrix}=LU \quad A=\begin{bmatrix}1&0&0\\1&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}1&1&0\\0&1&1\\0&0&1\end{bmatrix}=LDU$

(ii) $A=\begin{bmatrix}a&a&0\\a&a+b&b\\0&b&b+c\end{bmatrix} \xrightarrow{R_2-R_1} \begin{bmatrix}a&a&0\\0&b&b\\0&b&b+c\end{bmatrix} \xrightarrow{R_3-R_2} \begin{bmatrix}a&a&0\\0&b&b\\0&0&c\end{bmatrix} \to \begin{bmatrix}1&1&0\\0&1&1\\0&0&1\end{bmatrix}$

$\therefore U=\begin{bmatrix}a&a&0\\0&b&b\\0&0&c\end{bmatrix} \quad \therefore L=\begin{bmatrix}1&0&0\\1&1&0\\0&1&1\end{bmatrix} \quad D=\begin{bmatrix}a&0&0\\0&b&0\\0&0&c\end{bmatrix}$

$A=\begin{bmatrix}1&0&0\\1&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}a&a&0\\0&b&b\\0&0&c\end{bmatrix}=LU \quad A=\begin{bmatrix}1&0&0\\1&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}a&0&0\\0&b&0\\0&0&c\end{bmatrix}\begin{bmatrix}1&1&0\\0&1&1\\0&0&1\end{bmatrix}=LD\hat{U}$

8. $A=\begin{bmatrix}6&1\\3&2\end{bmatrix} \quad B=\begin{bmatrix}4&3\\1&2\end{bmatrix}$

$|A|=9 \quad |B|=5 \quad AB=\begin{bmatrix}25&20\\14&13\end{bmatrix} \quad |AB|=25\times13-14\times20=45$

$|A||B|=45=|AB| \quad \therefore \det(AB)=\det(A)\det(B)$

9. (i) $A=\begin{bmatrix}1&2\\2&-1\end{bmatrix}$ ① $|A|=-1-4=-5$ ② $A^{-1}=\frac{1}{-5}\begin{bmatrix}-1&-2\\-2&1\end{bmatrix}$

(ii) $A=\begin{bmatrix}3&1\\2&6\end{bmatrix}$ ① $|A|=18-2=16$ ② $A^{-1}=\frac{1}{16}\begin{bmatrix}6&-1\\-2&3\end{bmatrix}$

(iii) $A=\begin{bmatrix}3&3&1\\2&1&1\\-2&2&-1\end{bmatrix}$ ① $|A|=-3+4-6-(-2)-6+6=-3$ ② $A^{-1}=\frac{1}{-3}\begin{bmatrix}-3&5&2\\0&-1&-1\\6&-12&-3\end{bmatrix}$

(iv) $A=\begin{bmatrix}1&1&1\\0&2&1\\0&0&1\end{bmatrix}$ ① $|A|=2$ ② $A^{-1}=\frac{1}{2}\begin{bmatrix}2&-1&-1\\0&1&-1\\0&0&2\end{bmatrix}$

10. (i) $|A| = \begin{vmatrix} 4 & 0 & 0 & 0 \\ 7 & -1 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 7 & 2 & 1 & -3 \end{vmatrix} = 4(-1)^2(-1)3(-3) = 36$

(ii) $|A| = \begin{vmatrix} 7 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -6 & 5 & 7 & 2 \\ 2 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{vmatrix} = 2(-1)^{3+5} \begin{vmatrix} 7 & 3 & 2 & 4 \\ 9 & 0 & -4 & 1 \\ 2 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 \end{vmatrix} = 2 \times 2(-1)^{3+1} \begin{vmatrix} 3 & 2 & 4 \\ 0 & -4 & 1 \\ 2 & 3 & 2 \end{vmatrix} = 4 \times 3 = 12$

(iii) A is not square matrix, so $|A|$ don't exist.

(iv) A is not square matrix, so $|A|$ don't exist.

11. (i) $|AB| = |A||B| = 12$

(ii) $|5A| = 5^3|A| = 125 \times 3 = 375$

(iii) $|B^T| = |B| = 4$

(iv) $|A^{-1}| = \frac{1}{3}$

(v) $|A^3| = |AAA| = |A|^3 = 27$

12. (i) $|E| = |I_k B| = |I_k||B| = |B|$

(ii) $|F| = |A I_{n-k}| = |A||I_{n-k}| = |A|$

(iii) $|C| = |AB| = |A||B|$

13. (i) $\begin{vmatrix} a & 1 & 0 & 0 \\ -1 & b & 1 & 0 \\ 0 & -1 & c & 1 \\ 0 & 0 & -1 & d \end{vmatrix} = a \begin{vmatrix} b & 1 & 0 \\ -1 & c & 1 \\ 0 & -1 & d \end{vmatrix} + \begin{vmatrix} 1 & 0 & 0 \\ -1 & c & 1 \\ 0 & -1 & d \end{vmatrix} = a(bcd+b+d) + (cd+1) = abcd+ab+ad+cd+1$

(ii) $\begin{vmatrix} a & b & b & \cdots & b \\ b & a & b & \cdots & b \\ b & b & a & \cdots & b \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ b & b & b & \cdots & a \end{vmatrix} = \begin{vmatrix} a-b & 0 & 0 & \cdots & 0 \\ 0 & a-b & 0 & \cdots & 0 \\ 0 & 0 & a-b & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ b & b & b & \cdots & a \end{vmatrix} = (a-b)^{n-1} \begin{vmatrix} 1 & 0 & 0 & \cdots & -1 \\ 0 & 1 & 0 & \cdots & -1 \\ 0 & 0 & 1 & \cdots & -1 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ b & b & b & \cdots & a \end{vmatrix} = (a-b)^{n-1}[a+(n-1)b] = [a+(n-1)b](a-b)^{n-1}$

assume matrix is $n \times n$.

(iii) $A_n = (\alpha+\beta)A_{n-1} - \alpha\beta A_{n-2}$

$A_n - \alpha A_{n-1} = \beta(A_{n-1} - \alpha A_{n-2}) = \beta^2(A_{n-2} - \alpha A_{n-3}) = \cdots = \beta^{n-2}(A_2 - \alpha A_1) = \beta^{n-2}[(\alpha+\beta)^2 - \alpha\beta - \alpha(\alpha+\beta)] = \beta^n$

$\therefore A_n - \alpha A_{n-1} = \beta^n$

$\because \alpha$ and β are changeable $\therefore A_n - \beta A_{n-1} = \alpha^n$

$\therefore (\alpha-\beta)A_{n-1} = \alpha^n - \beta^n \quad \therefore (\alpha-\beta)A_n = \alpha^{n+1} - \beta^{n+1}$

$\therefore A_n = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha-\beta}$

14. $|A| = a_{11}a_{22} - a_{21}a_{12}$ $|B| = b_{11}b_{22} - b_{12}b_{21}$ $|C| = a_{11}b_{22} - b_{21}a_{12}$ $|D| = b_{11}a_{22} - a_{21}b_{12}$

$A+B = \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} \\ a_{21}+b_{21} & a_{22}+b_{22} \end{bmatrix}$ $|A+B| = (a_{11}+b_{11})(a_{22}+b_{22}) - (a_{12}+b_{12})(a_{21}+b_{21})$

$= a_{11}a_{22} + a_{11}b_{22} + b_{11}a_{22} + b_{11}b_{22} - a_{12}a_{21} - a_{12}b_{21} - b_{12}a_{21} - b_{12}b_{21}$

$\therefore |A+B| = |A| + |B| + |C| + |D|$

15. $\because I = U^T U \quad \therefore |I| = |U^T U| = |U^T||U| = 1$

$\because |U^T| = |U| \quad \therefore |U||U| = 1 \quad \therefore |U| = \pm 1$

16. (i) $|A| = \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & b-a & c-a & d-a \\ 0 & b^2-a^2 & c^2-a^2 & d^2-a^2 \\ 0 & b^3-a^3 & c^3-a^3 & d^3-a^3 \end{vmatrix} = \begin{vmatrix} b-a & c-a & d-a \\ b^2-a^2 & c^2-a^2 & d^2-a^2 \\ b^3-a^3 & c^3-a^3 & d^3-a^3 \end{vmatrix} = \begin{vmatrix} b-a & c-a & d-a \\ 0 & (c-b)(c-a) & (d-b) \\ 0 & (a+b+c)(c-a)(c-b) & (a+b+d)(d-a)(d-b) \end{vmatrix}$

$= (b-a)\left[(a+b+d)(c-a)(c-b)(d-a)(d-b) - (a+b+c)(c-a)(c-b)(d-a)(d-b)\right]$

$= (d-a)(c-a)(b-a)(d-b)(c-b)(d-c)$

(ii) If A is nonsingular $\Leftrightarrow |A| \neq 0$

$\because |A| = (d-a)(c-a)(b-a)(d-b)(c-b)(d-c)$ $\therefore \begin{cases} d \neq a \\ c \neq a \\ b \neq a \\ d \neq b \\ c \neq b \\ d \neq c \end{cases}$ $\therefore a \neq b \neq c \neq d$

$\therefore a, b, c, d$ are distinct.